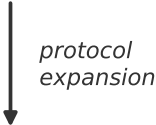


Protocol Hierarchy Expansion

Phase 1 of Stage 4 (4D Decomposition)

Original LRF Rules (first-match order)

r1:	TCP	192.168.1.5	ANY	443	→ ALLOW
r2:	IP	192.168.0.0/16	ANY	ANY	→ DENY



After Atomic Expansion (order preserved)

TCP:	192.168.1.5 / 443	→ ALLOW	(from r1)
TCP:	192.168.0.0/16 (rest)	→ DENY	(from r2)
UDP:	192.168.0.0/16	→ DENY	(from r2)
ICMP:	192.168.0.0/16	→ DENY	(from r2; type [0, 255])

$\Pi(\text{IP}) = \{\text{TCP}, \text{UDP}, \text{ICMP}\}$ (super-set: r2 yields three sub-rules)

$\Pi(\text{TCP}) = \{\text{TCP}\}$ (already atomic: r1 yields one sub-rule)

Sub-rules inherit the position of their parent rule, so first-match priority is preserved within each atomic protocol class before the sweep-line phase.